Aimee Schechter

aimee.schechter@colorado.edu Ph.D. Candidate / CU Boulder

Education

Master of Astrophysics, *University of Colorado Boulder* Dec 2021
Bachelor of Science, Astronomy, Honors, *The University of Texas at Austin* May 2019
Bachelor of Science, Physics, *The University of Texas at Austin* May 2019
Study Abroad, University of Sussex Spring 2017

Education

NSF Graduate Research Fellowship Program Honorable Mention Spring 2021
UT College of Natural Sciences Dean's Honored Graduate Spring 2019
UT College of Natural Sciences Distinction in Research Spring 2019
UT College of Natural Sciences Distinction in Service & Leadership Spring 2019
Department of Astronomy Karl G. Henize Memorial Scholarship Spring 2019
Undergraduate AAS Chambliss Award Honorable Mention Spring 2019
University of Texas Undergraduate Research Fellowship Fall 2018
John W. Cox Scholarship for Advanced Studies in Astronomy Summer 2018

Publications

Aimee L. Schechter and Caitlin Casey *Examining the Gas Outflow for a Typical Dusty Star-Forming Galaxy at z=2*, 2018 Res. Notes AAS 2 228
Stemo, A., Comerford, J., Barrows, R., Stern, D., Assef, R., Griffith, R., and **Schechter, A.** *A Catalog of 220 Offset and Dual AGNs: Increased AGN Activation in Major Mergers and Separations under 4 kpc*, ApJ, 923, 1, 2021

Selected Talks adn Posters

T: How to Identify Galaxy Mergers in CANDELS with IllustrisTNG50, Galaxies and AGN Seminar, STScI Sept 2021
T: Merging galaxies in HST and JWST: An interpretable suite of CNNs for identifying and understanding merger features from cosmic coffee hour to cosmic brunch, Hernquist Group Oct 2021
T: How to Identify Galaxy Mergers in CANDELS with IllustrisTNG50, Lorentz Center Workshop Mar 2021
P: Examining the Gas Outflow for a Typical Dusty Star-Forming Galaxy at *z*=2, Conference for Undergraduate Women in Physics Jan 2019
P: Examining the Gas Outflow for a Typical Dusty Star-Forming Galaxy at *z*=2, 233rd meeting of the American Astronomical Society Jan 2019
T: Examining the Gas Outflow for a Typical Dusty Star-Forming Galaxy at *z*=2, Texas Astronomy Undergraduate Research Symposium Oct 2018

Research Experience

CANDELS Merger Identification with IllustrisTNG50 **Jan 2020 — Present**
Mentored by Prof. Julie Comerford and Dr. Rebecca Nevin *CU Boulder*

- Identified merging galaxies using the Sublink merger trees
- Created a mock *HST* image pipeline for TNG50 galaxies
- Trained a test Convolutional Neural Network on TNG50 particle maps

Mapping Dust Emission in M101 to Improve Precision Cosmology **Oct 2018 — Dec 2019**
Mentored by Prof. Caitlin Casey *UT Austin*

- Corrected for dust along line of sight to Cepheid variable stars using dust emission maps
- Analyzed light re-emitted at longer wavelengths from dust
- Calculated correlation coefficients of Cepheid color and flux at long wavelengths

Examining the Gas Outflow for a Typical Dusty, Star-Forming Galaxy at *z*=2 **Aug 2017 — Sep 2018**
Mentored by Prof. Caitlin Casey *UT Austin*

- Extracted optical and infrared spectra in IRAF
- Used Python to analyze blueshift in absorption features
- Stacked absorption features and converted to velocity space to find outflow speed

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Analyzing the Cooling Tracks of Extremely Low Mass White Dwarfs **Aug 2016 — Dec 2016**
Mentored by Dr. Mike Montgomery *UT Austin*

- Simulated the accretion of an extremely low mass white dwarf from supernova remnant in MESA
- Plotted cooling track in Python
- Compared results to cooling tracks of extremely low mass white dwarfs formed by Roche-Lobe Overflow

Type II Supernovae Simulation **Mar 2016 — Jun 2016**
Mentored by Dr. Mike Montgomery *UT Austin*

- Simulated a 20 $M_\odot$ from main sequence through to core collapse using MESA
- Removed core and simulated supernova
- Analyzed mass fractions to find amount of ^{56}Ni synthesized

Teaching Experience

Graduate Teaching Certificate Expected Fall 2022
Lecture TA, ASTR 1200 Stars and Galaxies Spring 2020, Fall 2020
Lab TA ASTR 1010 Introductory Astronomy I Fall 2019

Outreach and Leadership

Lead Facilitator and Treasurer **Jul 2020 — Present**
Science Under the Dome, Fiske Planetarium *CU Boulder*

- Fund raise and handle finances for the organization
- Run auditions for future speakers
- Provide feedback for speakers on visuals, content, and public speaking

President **May 2017 — May 2018**
Undergraduate Women in Physics *UT Austin*

- Organized speakers and activities for weekly meetings
- Worked with department to improve undergraduate curriculum and department culture
- Created support system for underrepresented people in the department

Outreach Coordinator **Aug 2017 — Sep 2018**
Undergraduate Women in Physics *UT Austin*

- Planned events and coordinated volunteers
- Set up demos and explained complex physics to the public

Program Ambassador/Member **Aug 2015 — May 2019**
Women in Natural Sciences Program *UT Austin*

- Selected based on academic achievement and commitment to the community
- Represent the WINS Program at College events
- Tutored fifth grade student in reading comprehension and success strategies
- Mentored an incoming WINS student (August 2018-May 2019)

CU Astrophysical and Planetary Sciences Department Committees

Lead Graduate Representative, Friday Lunch Seminar Organizing Committee Fall 2021 — Present
Graduate Representative, Graduate Curriculum and Concerns Committee Fall 2021 — Present
Graduate Representative, Undergraduate Curriculum and Concerns Committee Fall 2019 — Spring 2021